About this PDF

Built only from: **NOC:Cloud Computing and Distributed Systems (NPTEL course book, weeks 1, 6)** — https://nptel.ac.in/courses/106104182

Licence: CC BY-SA 4.0. Accessed 2026-09-26.

Covers 13 of 13 syllabus sub-topics of this unit.

Partly covered in the source:

- Cloud Deployment Models
- Driving Factors and Challenges of Cloud
- Scalability and Elasticity

Definition of Cloud Computing

Cloud computing is a paradigm where software is delivered as a service to millions of users over the internet, rather than being distributed to run on individual PCs.

Explanation

The shift from distributed systems to cloud computing has been driven by applications with variable workloads and large data sets, leading to changes in machine architectures, operating systems, platforms, network connectivity and application workloads.

Virtualization of physical resources in large data centers creates a pool of virtualized resources that can be allocated to applications, providing flexibility and elasticity.

Cloud computing embodies the concept of utility computing, where computing resources are provided as a paid service, analogous to utilities like electricity or water.

In data-intensive computing, the focus shifts from CPU utilization to I/O and network throughput, as data cannot be moved easily and computation must occur near the data.

Example

Amazon Web Services, Microsoft Azure, and Google Compute Engine are prominent public cloud providers offering services such as compute, storage, and block storage.

Key points

- Cloud computing delivers software as a service to millions of users.
- Virtualization creates a pool of resources that can be allocated to applications, enabling elasticity.
- The evolution from distributed systems to cloud computing is driven by variable workloads and large data sets.
- Data-intensive computing prioritizes I/O and network performance over CPU usage.

- Public clouds are accessible to paying customers, while private clouds are managed internally by a single organization.

Characteristics of Cloud

Definition

Characteristics of Cloud refer to the features that distinguish cloud computing from previous generations. These features include massive scale, on demand access, and data intensive computation.

Explanation

Massive scale means that very large data centers contain tens of thousands of servers and applications can run across as many servers as needed. On demand access means pay-as-you-go, contrasting with upfront cost. Data intensive nature refers to problems that involve very large data sizes such as terabytes, petabytes, and zettabytes, which are common in daily logs, forensics reports, and weblogs.

Example

An example of a data intensive problem is the continuous generation of weblogs that grow to very large sizes and must be processed.

Key points

- massive scale
- on demand access
- data intensive nature

Cloud Deployment Models

Partly covered in the source.

Definition

Geographical distributed clouds are a deployment model in which cloud providers deploy data centers at multiple sites.

Explanation

In this model, each site may have a different structure and the services running within it can communicate over a fast network.

Key points

- Multiple sites are used.
- Sites are connected over a fast network.
- Each site may have a different structure.

Cloud Service Models

Definition

Cloud service models are categories of services offered by cloud providers that allow customers to use computing resources on demand. These models include infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS).

Explanation

The service models are built on the principle of on-demand access, where customers pay only for the resources they use. Virtualization enables the abstraction of physical hardware, allowing providers to offer flexible computing and storage infrastructure (IaaS). PaaS couples this infrastructure with a programming platform, providing a ready-to-use environment for application development. SaaS delivers complete software applications over the internet, subsuming service-oriented architectures. These models support the utility computing paradigm, where computing is treated like a utility that can be plugged in and used as needed.

Example

- **AWS EC2** – an example of infrastructure as a service, allowing users to rent virtual servers and pay per CPU hour.
- **AWS S3** – an example of infrastructure as a service, providing storage that is billed per gigabyte per month.
- **Google App Engine** – an example of platform as a service, offering a managed environment for running applications.
- **Google Docs** – an example of software as a service, delivering a document editor over the web.
- **Microsoft Office on Demand** – another example of software as a service.

Key points

- Cloud service models are categorized as IaaS, PaaS, and SaaS.
- They rely on on-demand access and pay-as-you-go pricing.
- Virtualization underpins the delivery of these services.
- Examples include AWS EC2, AWS S3, Google App Engine, Google Docs, and Microsoft Office on Demand.

Driving Factors and Challenges of Cloud

Partly covered in the source.

Explanation

Cloud computing data centres consist of servers housed in racks, each server containing blades or boards that are powered and network-connected. The operation of these servers requires a large amount of electrical power, which is supplied by power stations. The high power consumption generates significant heat, necessitating cooling solutions. Cooling is achieved through various methods, such as air intake combined with purified water that carries the heat away. The efficiency of these systems is measured by metrics such as annual water usage per unit of IT equipment energy and the power utility ratio (total facility power divided by IT equipment power). Lower values of these metrics indicate better energy efficiency. Google’s data centre demonstrates a power usage effectiveness (PUE) of 1.11, which is close to the theoretical minimum of 1, indicating minimal energy wastage in computing operations.

Example

Google’s data centre achieves a PUE of 1.11, showing that it uses almost all the power drawn for computing purposes with minimal waste.

Key points

- Data centres require large amounts of power, leading to significant heat generation.
- Cooling is essential and can involve air and water systems to manage heat.
- Energy efficiency is evaluated using metrics like water usage per IT energy and power utility ratio.
- A lower PUE value indicates better energy efficiency; Google’s data centre has a PUE of 1.11.

Virtualization

Definition

Virtualization is the process of creating virtualized resources from physical resources, forming a pool of virtual resources that can be allocated to applications.

Explanation

Virtualization enables the cloud to build on physical or virtualized resources over large distributed data centers, providing flexibility and elasticity in computing resources. It allows infrastructure as a service to be delivered by abstracting hardware, enabling on-demand access to computing and storage.

Example

AWS Elastic Compute Cloud (EC2) demonstrates virtualization by provisioning a new virtual server within minutes, reducing the time and cost compared to purchasing and installing physical hardware.

Key points

- Virtualization creates a pool of virtual resources from physical hardware.
- It underpins infrastructure as a service, allowing on-demand access to computing and storage.
- Cloud providers like AWS use virtualization to rapidly deploy virtual servers, saving time and operational costs.

Load Balancing

Definition

Load balancing refers to the distribution of data regions across multiple region servers to evenly spread workload and provide fault tolerance.

AcademicsGen — source: NPTEL — CC BY-SA 4.0

Explanation

Auto sharding splits a table into regions, each covering a continuous range of row keys. These regions are stored on region servers and are moved around for load balancing and failover. The master cluster coordinates the distribution, while Zookeeper stores control information and helps clients locate the appropriate region server.

Example

A table with row keys A–S can be split into three regions: A–F, G–L, and M–S. The first region (A–C) and the last range (T–Z) may both be stored on region server 1, illustrating how splits are distributed across servers.

Algorithm / Steps

1. Master splits the table into regions based on row key ranges.
2. Regions are assigned to region servers in a random order.
3. Master monitors the number of regions per server and moves regions to new or under-utilized servers to balance load.
4. Zookeeper stores the coordination and control information for the cluster.
5. Clients use the master or Zookeeper to locate the region server that holds the desired region and access data directly.

Key points

- Regions are horizontal partitions of a table defined by row key ranges.
- Auto sharding allows automatic or manual splitting to scale with data growth.
- Load balancing is achieved by moving regions among region servers as the cluster expands.
- Zookeeper acts as a coordination service storing control information.
- Clients access data through region servers identified via the master or Zookeeper.

Scalability and Elasticity

Partly covered in the source.

Definition

Scalability refers to the ability of a computing system to handle increasing workloads by adding resources such as storage, bandwidth, or processing power. Elasticity is the flexibility of a cloud system to dynamically allocate and release computing resources as demand changes. ,

Explanation

Scalable computing over the internet has evolved with variable workloads and large data sets, requiring changes in machine architectures, operating systems, platforms, and network connectivity. The cloud’s elasticity allows resources to be provisioned on demand, enabling users to pay only for what they use. ,

Scalability trends are evident in hardware growth: storage capacity is expected to double every 12 months, bandwidth every 9 months, and CPU compute capacity every 18 months, following a doubling phenomenon similar to Moore's law.

Example

Amazon EC2 (Elastic Compute Cloud) lets users run arbitrary operating system images and pay per CPU hour used, demonstrating on-demand access and elasticity.

Key points

- Cloud systems are designed to scale with variable workloads and large data sets.
- Elasticity provides dynamic resource allocation, allowing pay-as-you-go usage.
- Hardware growth follows a doubling pattern: storage (12 mo), bandwidth (9 mo), CPU (18 mo).
- Amazon EC2 exemplifies elastic compute resources with per-CPU-hour billing.

Replication

Replication is the process of storing multiple copies of data across different servers to ensure availability and fault tolerance.

Definition

Replication is the process of storing multiple copies of data across different servers.

Explanation

Replication allows large tables to be distributed across multiple region servers, improving accessibility and fault tolerance. In HDFS, the default replication factor is 3, meaning each block is stored on three different nodes. In cross-data-centre replication, the master cluster synchronously sends HLogs to the slave cluster, and Zookeeper coordinates the clusters.

Example

In HDFS, the default replication factor is 3, so each block is stored on three different nodes.

Algorithm / Steps

1. The master cluster synchronously sends HLogs to the slave cluster.
2. Zookeeper coordinates the clusters.
3. The slave cluster stores the received logs.

Key points

- Replication stores multiple copies of data across servers.
- HDFS default replication factor is 3.
- Cross-data-centre replication uses HLogs and Zookeeper for coordination.
- Replication improves accessibility and fault tolerance.

Monitoring

Definition

Monitoring in this context refers to ensuring that all data nodes and the master are alive and available at all times.

Explanation

Zookeeper is used for distributed coordination; it ensures that all data nodes are alive and the master is available. The HBase master monitors all region server instances in the cluster.

Example

HBase master monitors all region server instances in the cluster.

Algorithm / Steps

1. Zookeeper ensures all data nodes are alive and the master is available.
2. HBase master monitors all region server instances in the cluster.

Key points

- Zookeeper provides distributed coordination and liveness checks for data nodes and the master.
- The HBase master actively monitors region server instances.

Cloud Services and Platforms: Compute Services

Definition

Compute services provide virtualized computing resources over the internet, allowing users to rent hardware and software infrastructure on demand.

Explanation

Compute services are delivered through virtualization, which abstracts physical hardware into virtual machines. Users can provision and manage these virtual machines without owning the underlying hardware. The services are offered on an on-demand basis, meaning customers pay only for the resources they consume, typically measured in CPU hours or storage GB per month. The model is classified as infrastructure as a service (IaaS), platform as a service (PaaS), or software as a service (SaaS) depending on the level of abstraction provided.

Example

Amazon Web Services offers Elastic Compute Cloud (EC2) for virtual machine provisioning and Simple Storage Service (S3) for object storage. Google App Engine provides a platform for deploying applications without managing infrastructure. Microsoft Azure offers similar IaaS and PaaS offerings.

Algorithm / Steps

1. Request the desired computing or storage resource from the provider.
2. Pay for the resource based on actual usage (e.g., CPU hours, GB per month).
3. Use the resource to run applications or store data.

4. Release the resource when no longer needed, ending the billing.

Key points

- Virtualization abstracts physical hardware into virtual machines, enabling flexible provisioning.
- On-demand access allows users to pay only for the resources they consume, eliminating upfront costs.
- Compute services are classified as IaaS, PaaS, or SaaS based on the level of abstraction.
- Major cloud providers such as Amazon Web Services, Microsoft Azure, and Google Cloud offer compute services like EC2, Azure VMs, and App Engine.
- Compute services support massive scale and data-intensive workloads by providing thousands of servers and high I/O throughput.

Storage Services

Definition

Storage services provide on-demand access to storage resources, typically billed per GB per month.

Explanation

Storage services allow customers to rent storage rather than purchase hardware, paying only for the space they use. The provider maintains the infrastructure, including storage backend nodes equipped with SSDs that store data on disk. Data can be organized in tables, split into regions, and stored on a distributed file system such as HDFS. The system is coordinated by services like Zookeeper to ensure fault tolerance.

Example

AWS Simple Storage Service (S3) lets users store arbitrary datasets and pay per GB per month for the space used.

Key points

- Storage services are on-demand and billed per GB per month.
- Providers maintain the infrastructure, including SSD-based storage backend nodes.
- Data can be organized into tables, regions, and column families, and stored on HDFS.
- NoSQL databases like HBase and Cassandra offer different trade-offs between consistency, availability, and partition tolerance.

Application Services

Definition

Application services are cloud offerings that allow users to develop, run, and use applications without managing underlying infrastructure. They provide a programming platform or ready-made software that can be accessed on demand.

Explanation

These services combine flexible computing and storage infrastructure with a programming platform or software stack. The programming paradigm is supplied by the provider, enabling users to focus on application logic rather than hardware or operating system details. The services are typically pay-as-you-go, eliminating upfront costs.

Example

Google App Engine lets developers build and deploy applications within a managed framework, and Google Docs or Microsoft Office on demand are examples of software-as-a-service offerings that provide ready-to-use applications.

Algorithm / Steps

1. Obtain access to the cloud platform or software service.
2. Use the provided programming paradigm or software interface to develop or configure the application.
3. Deploy or run the application on the platform, paying only for the resources consumed.

Key points

- Application services bundle computing, storage, and a programming or software stack.
- They enable users to run applications without managing hardware or operating systems.
- Examples include Google App Engine, Google Docs, and Microsoft Office on demand.

Source and licence

This PDF is adapted from **NOC:Cloud Computing and Distributed Systems (NPTEL course book, weeks 1, 6)** (https://nptel.ac.in/courses/106104182), accessed 2026-09-26.

Source licence: **CC BY-SA 4.0** — https://creativecommons.org/licenses/by-sa/4.0/

Changes made by AcademicsGen: the source text was rephrased, reorganised and grammar-corrected into study material; nothing added.

This PDF is shared under the same licence (CC BY-SA 4.0).